

EES301: Global biogeochemical cycles and Climate change

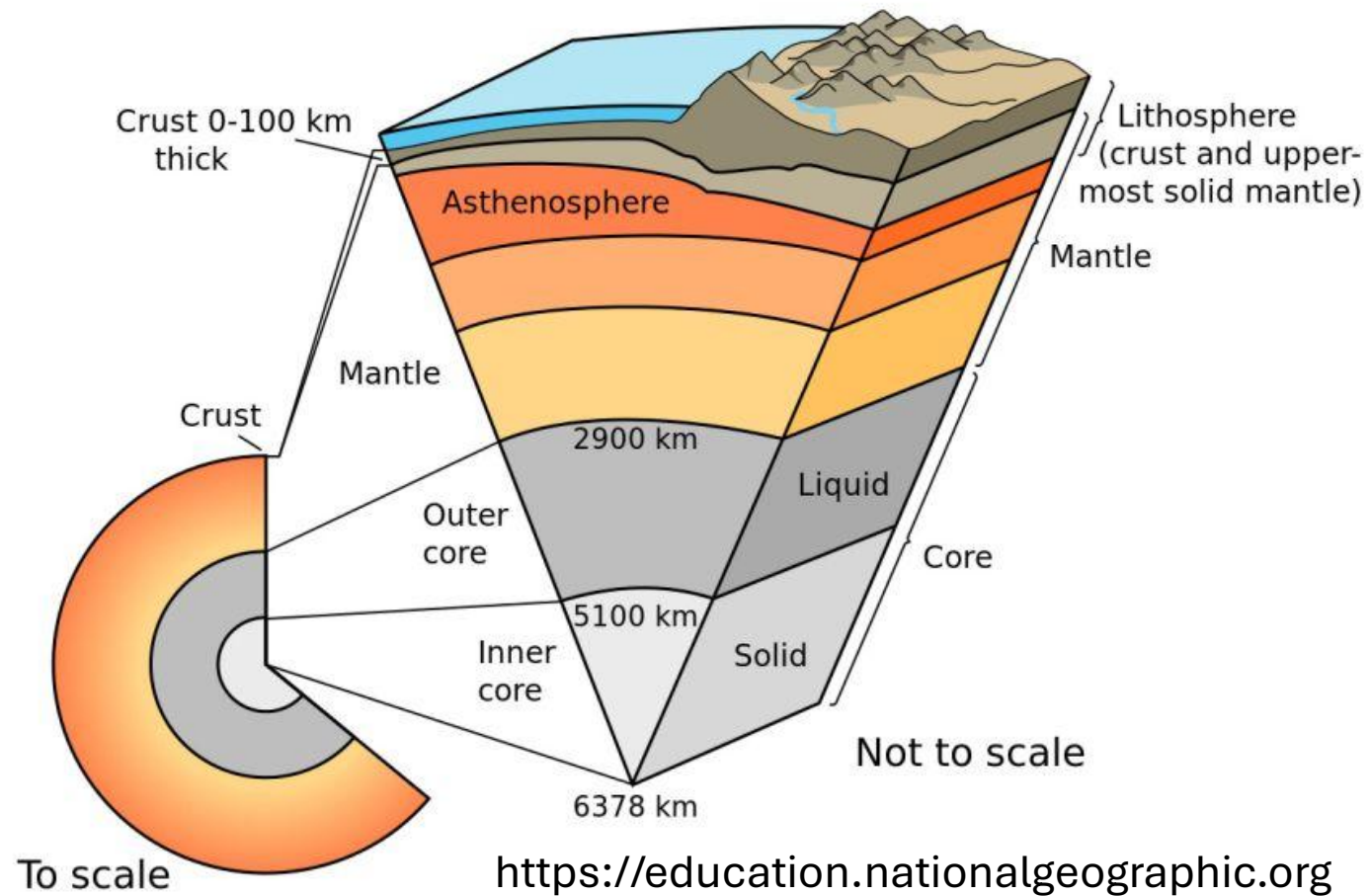
Properties of key reservoirs and mode
of transfers between key reservoirs

Lithosphere: Soils

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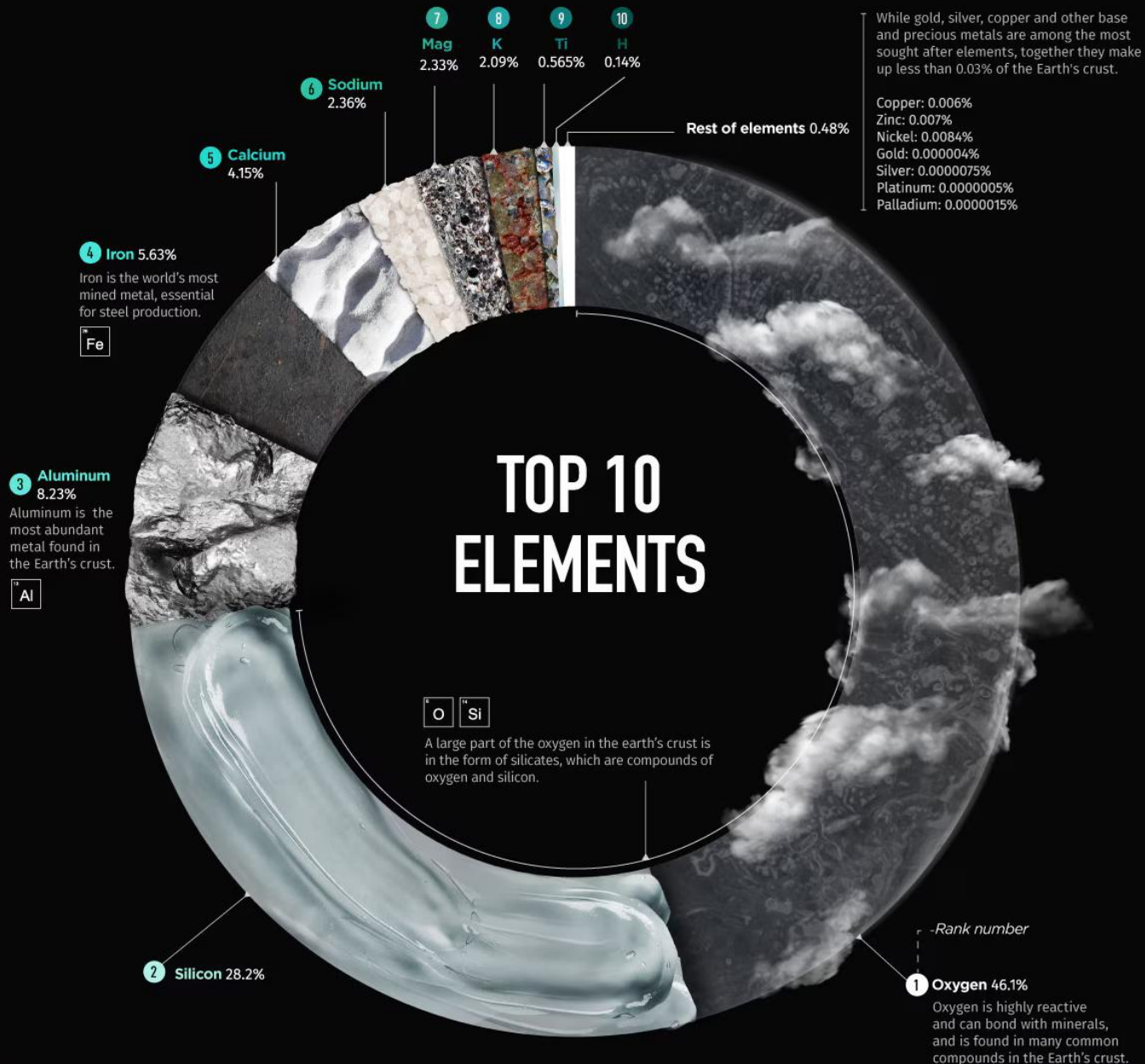
Lithosphere

- Solid rocky outer layer of Earth comprising the crust and upper mantle - the crust makes up to 1% of the planet's volume
- Oceanic and continental crust has distinct chemical compositions
- Contains vast mineral resources and serves as a long-term storage for elements
- Comprises about 95% igneous and metamorphic rocks, 4% shale, 0.75% sandstone, and 0.25% limestone



- Undergoes constant changes through tectonic processes, **weathering**, and metamorphism
- Interacts with other reservoirs through volcanic activity, weathering, and sediment deposition (erosion)

The Earth's crust is only 1% of the planet's volume but it contains the materials we use everyday.
Here is the abundance of elements in the Earth's crust by percentage (%).



Abundance of elements in Earth's crust

Element and Symbol	Percentage by weight	Percentage by number of atoms	Percentage by volume
Oxygen (O)	46.6	62.6	93.8
Silicon (Si)	27.7	21.2	0.9
Aluminum (Al)	8.1	6.5	0.5
Iron (Fe)	5.0	1.9	0.4
Calcium (Ca)	3.6	1.9	1.0
Sodium (Na)	2.8	2.6	1.3
Potassium (K)	2.6	1.4	1.8
Magnesium (Mg)	2.1	1.9	0.3
All other elements	1.5		
	100	100	100

[sites.tntech.edu/hwleimer/geol-1040](https://www.weforum.org)

<https://www.weforum.org>

Hydrogen 0.15%	Niobium 0.0017%	Thallium 0.000054%
Helium 5.5×10 ⁻⁷ %	Molybdenum 0.00011%	Lead 0.001%
Lithium 0.0017%	Technetium 0%	Bismuth 2.5×10 ⁻⁶ %
Beryllium 0.00019%	Ruthenium 1×10 ⁻⁷ %	Polonium N/A
Boron 0.00086%	Rhodium 7×10 ⁻⁸ %	Astatine 0%
Carbon 0.18%	Palladium 6.3×10 ⁻⁷ %	Radon N/A
Nitrogen 0.002%	Silver 7.9×10 ⁻⁶ %	Francium N/A
Oxygen 46%	Cadmium 0.000015%	Radium 1×10 ⁻¹¹ %
Fluorine 0.054%	Indium 0.000016%	Actinium N/A
Neon 3×10 ⁻⁷ %	Tin 0.00022%	Thorium 0.0006%
Sodium 2.3%	Antimony 0.00002%	Protactinium 9.9×10 ⁻¹³ %
Magnesium 2.9%	Tellurium 1×10 ⁻⁷ %	Uranium 0.00018%
Aluminum 8.1%	Iodine 0.000049%	Neptunium 0%
Silicon 27%	Xenon 2×10 ⁻⁹ %	Plutonium 0%
Phosphorus 0.099%	Cesium 0.00019%	Americium 0%
Sulfur 0.042%	Barium 0.034%	Curium 0%
Chlorine 0.017%	Lanthanum 0.0034%	Berkelium 0%
Argon 0.00015%	Cerium 0.006%	Californium 0%
Potassium 1.5%	Praseodymium 0.00086%	Einsteinium 0%
Calcium 5%	Neodymium 0.0033%	Fermium 0%
Scandium 0.0026%	Promethium 0%	Mendelevium 0%
Titanium 0.66%	Samarium 0.0006%	Nobelium 0%
Vanadium 0.019%	Europium 0.00018%	Lawrencium 0%
Chromium 0.014%	Gadolinium 0.00052%	Rutherfordium 0%
Manganese 0.11%	Terbium 0.000093%	Dubnium 0%
Iron 6.3%	Dysprosium 0.00062%	Seaborgium 0%
Cobalt 0.003%	Holmium 0.00012%	Bohrium 0%
Nickel 0.009%	Erbium 0.0003%	Hassium 0%
Copper 0.0068%	Thulium 0.000045%	Meitnerium 0%
Zinc 0.0078%	Ytterbium 0.00028%	Darmstadtium 0%
Gallium 0.0019%	Lutetium 0.000057%	Roentgenium 0%
Germanium 0.00014%	Hafnium 0.00033%	Copernicium 0%
Arsenic 0.00021%	Tantalum 0.00017%	Nihonium 0%
Selenium 5×10 ⁻⁶ %	Tungsten 0.00011%	Flerovium 0%
Bromine 0.0003%	Rhenium 2.6×10 ⁻⁷ %	Moscovium 0%
Krypton 1.5×10 ⁻⁸ %	Osmium 1.8×10 ⁻⁷ %	Livermorium 0%
Rubidium 0.006%	Iridium 4×10 ⁻⁸ %	Tennessine 0%
Strontium 0.036%	Platinum 3.7×10 ⁻⁶ %	Oganesson 0%
Yttrium 0.0029%	Gold 3.1×10 ⁻⁷ %	
Zirconium 0.013%	Mercury 6.7×10 ⁻⁶ %	

- Most elements are found in only trace amounts within crust - O, Si, Al, and Fe account for ~88 %, while another 90 elements make up the remaining ~12%.
- C, N, S, and P account for only about ~0.0323%.
- Ag, Au, Cu, etc. are the most sought-after elements, but they make up <0.03% of the Earth's crust by mass.
- O and Si are fundamental to most minerals in the Earth's crust, which are typically compounds rather than elemental forms.

Top 5 elements

- **Oxygen** – a highly reactive element that combines with other elements, forming oxides - granite and quartz (oxides of Si), rust (oxides of Fe), and limestone (oxide of Ca and C)
- **Silicon** – links up with oxygen to form the most common minerals - sand primarily consists of silica (silicon dioxide), usually in the form of quartz; >90% Earth has silicate minerals
- **Aluminum** – has strong affinity for oxygen, thus rarely found in its elemental state; E.g., Aluminum oxide (Al_2O_3), aluminum hydroxide ($\text{Al}(\text{OH})_3$) and potassium aluminum sulphate ($\text{KAl}(\text{SO}_4)_2$)
- **Iron** – minerals include hematite (Fe_2O_3) and magnetite (Fe_3O_4) – an essential nutrient for life forms
- **Calcium** – minerals include limestone (CaCO_3), gypsum (CaSO_4) and fluorite (CaF_2)

Weathering

- Following geologic uplift and exposure at the Earth's surface, rocks undergo weathering through a wide variety of processes that breakdown rocks and minerals.
- Crucial role in shaping the Earth's landscape over long periods of time.
- Important part of the rock cycle and contributes to the formation of soils.
- Major types:
 - **Physical/mechanical weathering**
 - **Chemical weathering**



Top:
Weathering
of granite



Bottom:
Limestone
dissolution
by carbonate
reaction -
honeycomb
pattern.

<https://ugc.berkeley.edu/background-content/weathering/>

Mechanical/Physical weathering

- Involves changes in the degree of consolidation/strength with little or no chemical and mineralogical changes of rocks.

Freeze-Thaw Weathering or Frost Wedging

expansion of frozen water in rocks leading to fracturing of rocks in cold climates

Exfoliation Weathering or Unloading Due to Thermal Stress

thermal expansion of rocks in arid/desert regions

Salt Weathering or Haloclasty

growth of salt crystals in crevices can force open fractures

Wind/Air abrasion

- **Biological weathering** -
Plant roots, endolithic
algae, lichens, etc. growing
in rock crevices



<http://www.gdargaud.net/>

- Physical weathering is important in exposing the minerals in rocks to chemical attack - if the rate is slow, it may limit the rate of chemical weathering and lower the fertility of soils

Chemical weathering

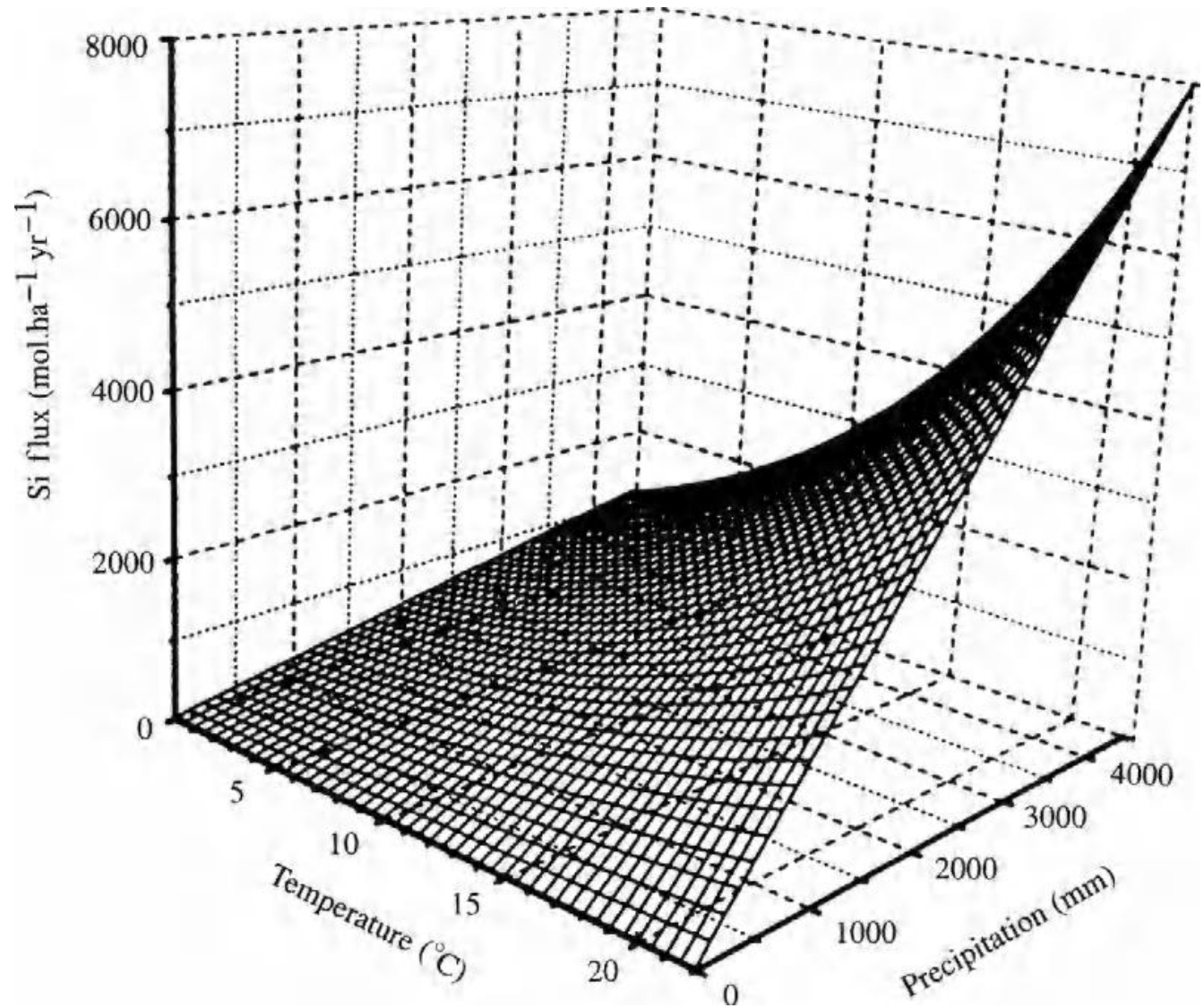
- Involves changes in chemical and mineralogical composition in rocks when the minerals in rocks and soils react with chemical substances.
- **The process/rate depends on**
 - **mineralogy** (primarily mineral composition of rocks; solubility - number and strength of the chemical bonds within the crystal lattice)
 - **climate** (e.g., rapidly under conditions of higher temperature and precipitation)
 - **composition of the soil solution** (pH, acids, redox, complexing ligands, etc.)
 - **water** (not only solvent but also means for redistribution of weathering products)
 - **tectonics** (movement of tectonic plates that shapes the Earth's crust)
 - **biological processes** (plants and soil microbes)

- Chemical weathering is more rapid in tropical forests >> temperate forests, most forests >> grasslands or deserts, and at low rates in cold environments

- Loss of Si in stream water is directly related to precipitation and temperature – an index of chemical weathering.

- **Processes:**

Dissolution, Hydration,
Hydrolysis, Acidolysis,
Chelation, Oxidation /
Reduction



Loss of silicon (SiO_2) in runoff as a function of mean annual temperature and precipitation in various areas of the world. Source: White and Blum (1995)

Dissolution:

- Minerals break down and separate into their component ions in water – salts, sulfates and carbonates

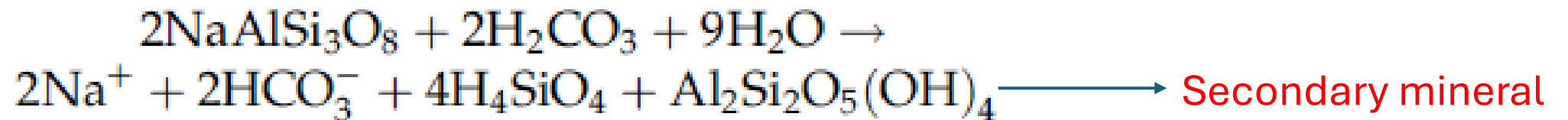
E.g., Calcite dissolution: $\text{CaCO}_3 \rightarrow \text{Ca}^{2+} + \text{CO}_3^{2-}$

- Carbonation reaction** - driven by the formation of carbonic acid, H_2CO_3 - the most dominant process



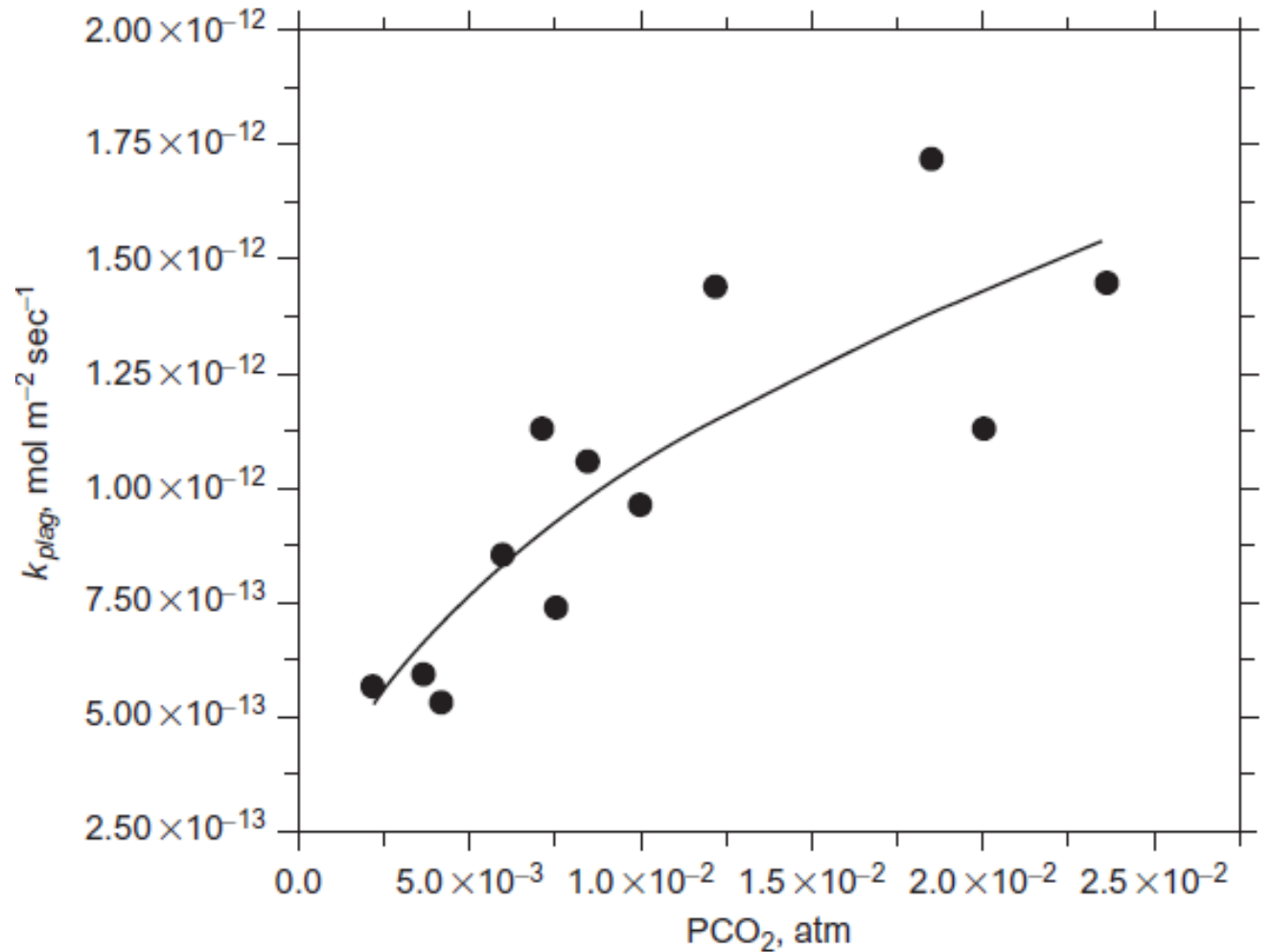
Congruent reaction –
entire material is
weathered into
soluble products

E.g. weathering of Na-feldspar, albite (**Incongruent reaction**)



Carbonate weathering – climate change impact

- It consumes CO_2 - hence as atmospheric CO_2 increases, the rates of rock weathering may also increase.
- **Increased soil CO_2 concentrations** - the amount of CO_2 in the soil profile increases when plants are grown at high CO_2 and due to high microbial activity



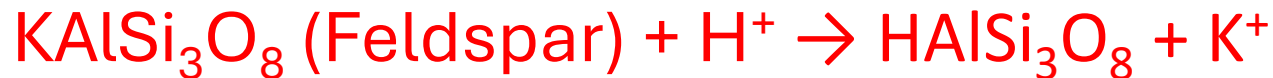
Dissolution of Ca-feldspar (plagioclase) as a function of soil CO_2 concentrations in watersheds of the Sierra Nevada (California), subject to differential hydrothermal activity.
Source: From Navarre-Sitchler and Thyne (2007).

Hydration and Hydrolysis

- **Hydration** is the incorporation of water molecules into a mineral, weakening the stability of the mineral, making it susceptible to other forms of weathering.



- **Hydrolysis** is the incorporation of either H^+ or OH^- (from water) into a mineral.



Acidolysis

- Similar to hydrolysis in that H^+ is incorporated into a mineral, but the source is organic or inorganic acids, such as oxalic acid, acetic acid, fulvic acid, carbonic acid, nitric/sulfuric acid, etc.

Chelation

- Organic acids can cause chelation – a ligand capable of forming multiple bonds with a metal ion such as Fe, Al or Ca in minerals (e.g., $AlPO_4$), resulting in the metal incorporated into the complex and stripping of metal ions from primary minerals.

Oxidation and Reduction

- Weathering of minerals by transfer of electrons; Fe, Mn, S -containing minerals are susceptible to redox reactions.

E.g., Fe reduction and oxidation

Fe^{2+} (insoluble) oxidation $\rightarrow$ Fe^{3+} (soluble) (oxic condition)



Pyrite

Hematite

Fe^{3+} (insoluble) reduction $\rightarrow$ Fe^{2+} (soluble) (anoxic condition)

Roles of life forms in weathering processes

- Plant roots and soil microbes release CO_2 to the soil - **By maintaining high concentrations of CO_2 in the soil, they control the process of rock weathering** – Carbonate dissolution
- Microbes release/produce **organic acids** (e.g., acetic/ citric/ fluvic/ humic acid) - Acidolysis
- Microbes are also involved in **oxidation/reduction reactions**.

All these are examples of biogeochemistry at the Earth's surface

Table 8-1 Primary and secondary minerals commonly found in soils

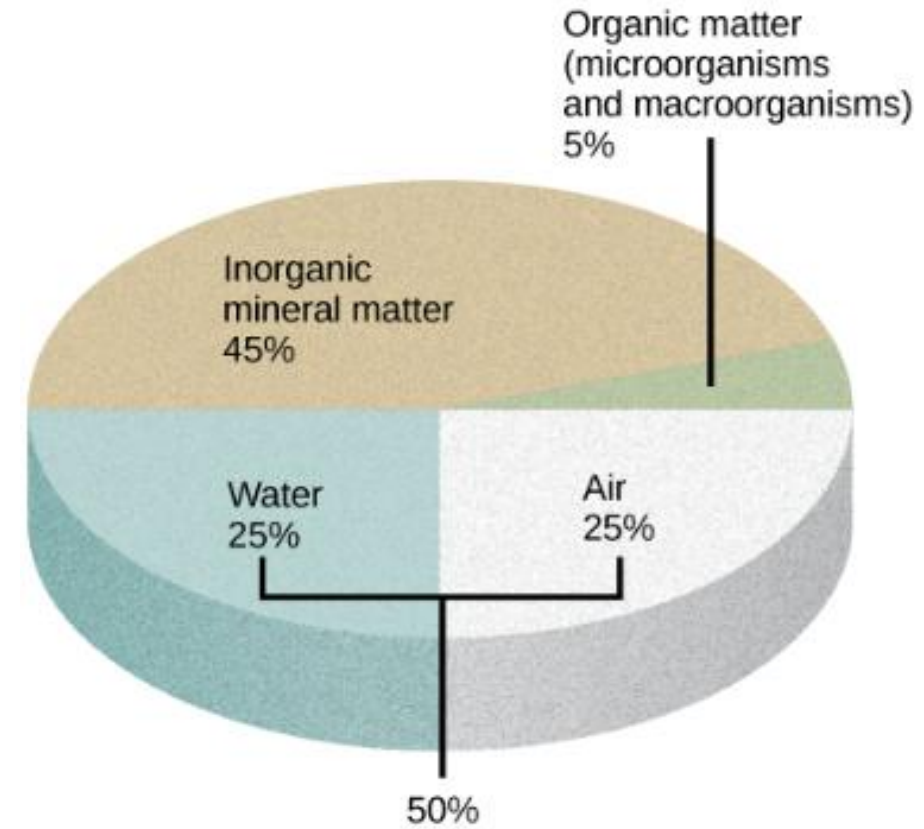
Primary minerals	Approximate composition	Weatherability
Quartz	SiO_2	—
K-Feldspar	KAlSi_3O_8	+
Ca, Na-plagioclase	$\text{CaAl}_2\text{Si}_2\text{O}_8$ to $\text{NaAlSi}_3\text{O}_8$	+ to (+)
Muscovite	$\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2$	+(+)
Amphibole	$\text{Ca}_2\text{Al}_2\text{Mg}_2\text{Fe}_3\text{Si}_6\text{O}_{22}(\text{OH})_2$	+(+)
Biotite	$\text{KAl}(\text{Mg,Fe})_3\text{Si}_3\text{O}_{10}(\text{OH})_2$	++
Pyroxene	$\text{Ca}_2(\text{Al,Fe})_4(\text{Mg,Fe})_4\text{Si}_6\text{O}_{24}$	++
Apatite	$[\text{3Ca}_3(\text{PO}_4)_2] \cdot \text{CaO}$	++
Volcanic glass	Variable	++
Calcite	CaCO_3	+++
Dolomite	$(\text{Ca,Mg})\text{CO}_3$	+++
Gypsum	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	+++
Secondary minerals	Approximate composition	Type
Kaolinite	$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$	1:1 layer-silicate
Vermiculite	$(\text{Al}_{1.7}\text{Mg}_{0.3})\text{Si}_{3.6}\text{Al}_{0.4}\text{O}_{10}(\text{OH})_2$	2:1 layer-silicate
Montmorillonite	$(\text{Al}_{1.7}\text{Mg}_{0.3})\text{Si}_{3.9}\text{Al}_{0.1}\text{O}_{10}(\text{OH})_2$	2:1 layer-silicate
Chlorite	$(\text{Mg}_{2.6}\text{Fe}_{0.4})\text{Si}_{2.5}(\text{Al,Fe})_{1.5}\text{O}_{10}(\text{OH})_2$	2:1:1 layer-silicate
Allophane	$(\text{SiO}_2)_{1-2}\text{Al}_2\text{O}_5 \cdot 2.5-3(\text{H}_2\text{O})$	Pseudocrystalline, spherical
Imogolite	$\text{SiO}_2\text{Al}_2\text{O}_3 \cdot 2.5\text{H}_2\text{O}$	Pseudocrystalline, strands
Hallyosite	$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4 \cdot 2\text{H}_2\text{O}$	Pseudocrystalline, tubular
Gibbsite	$\text{Al}(\text{OH})_3$	Hydroxide
Goethite	FeOOH	Oxyhydroxide
Hematite	Fe_2O_3	Oxide
Ferrihydrite	$5\text{Fe}_2\text{O}_5 \cdot 9\text{H}_2\text{O}$	Oxide

Weathering and erosion of rocks affects Earth's surface features (i.e., topography), soil, nutrients distribution (land/oceans), and the atmospheric composition, which in turn affects global climate and ecosystems

Weathering	Erosion
Breaking down/decomposition of rocks, minerals and soils	Displacement of solids/soils by air/water/ice/gravity
Static process	Dynamic process
Play important role in soil formation	Destructive process – soil losses and transport
Physical/chemical/biological processes involved	Water/wind/snow/anthropogenic processes involved

Soils

- Multi-phase system consisting of solids, liquids and gases
- **Soil solids:** Inorganic - mineral matter
Organic – detritus and living matter
- **Soil air** differs from the atmosphere due to microbial activity/ecology – more CO_2 (1 – 10%) and less O_2 (5-20%) – depends on porosity
- **Soil liquid/water/solution** – interface between soil phases, providing transport mechanisms; can lead to leaching of elements and soil erosion



Brock Biology of Microorganisms, 15th ed.

Pedosphere: the layer/envelope of the Earth where soils occur and where soil-forming processes are active

Soil formation and layers

- Influenced by climate, time, biota, topographic position, and the parent material of the site.
- **Fundamental processes:** Inputs, Outputs, Transformation and Translocation
- Soils usually consist of a number of layers, or horizons - collectively constitute the **soil profile, or Pedon**
- Rock weathering, water movement, and organic decomposition influence the development of a soil profile under varying climatic conditions

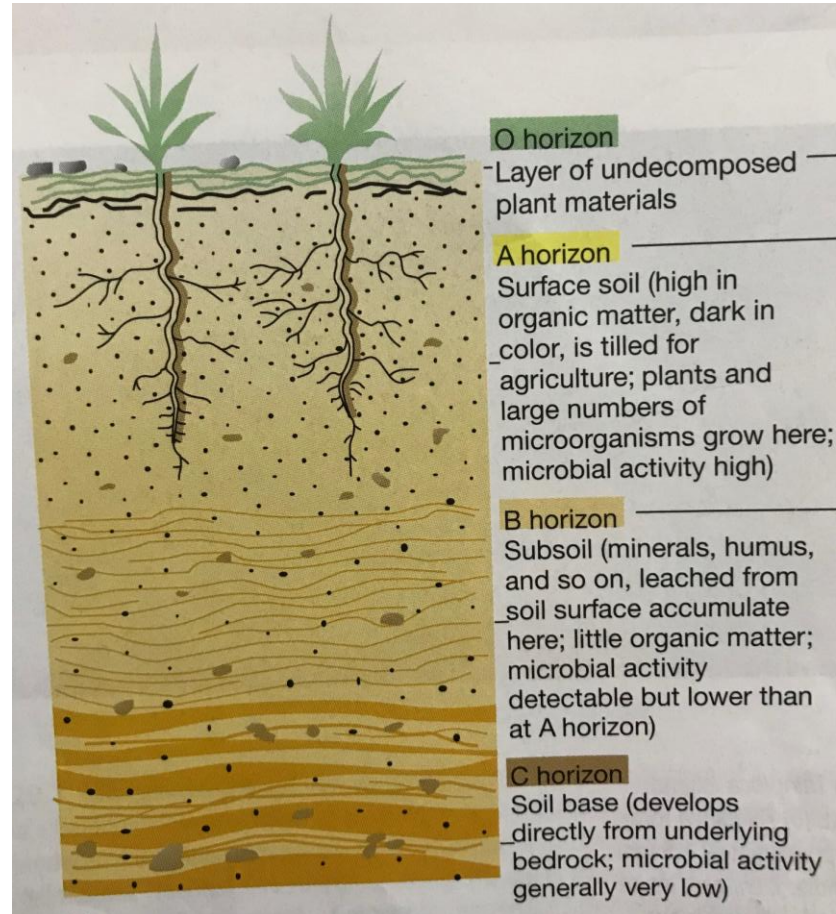
Soil horizons/layers

differ in

composition (e.g.,
organic matter/clay
content),

physical properties (e.g.,
color or particle size) or

chemical properties
(e.g., pH or conductivity)



Brock Biology of Microorganisms, 15th ed.

Rock cycle

Uplift

Weathering

or Erosion

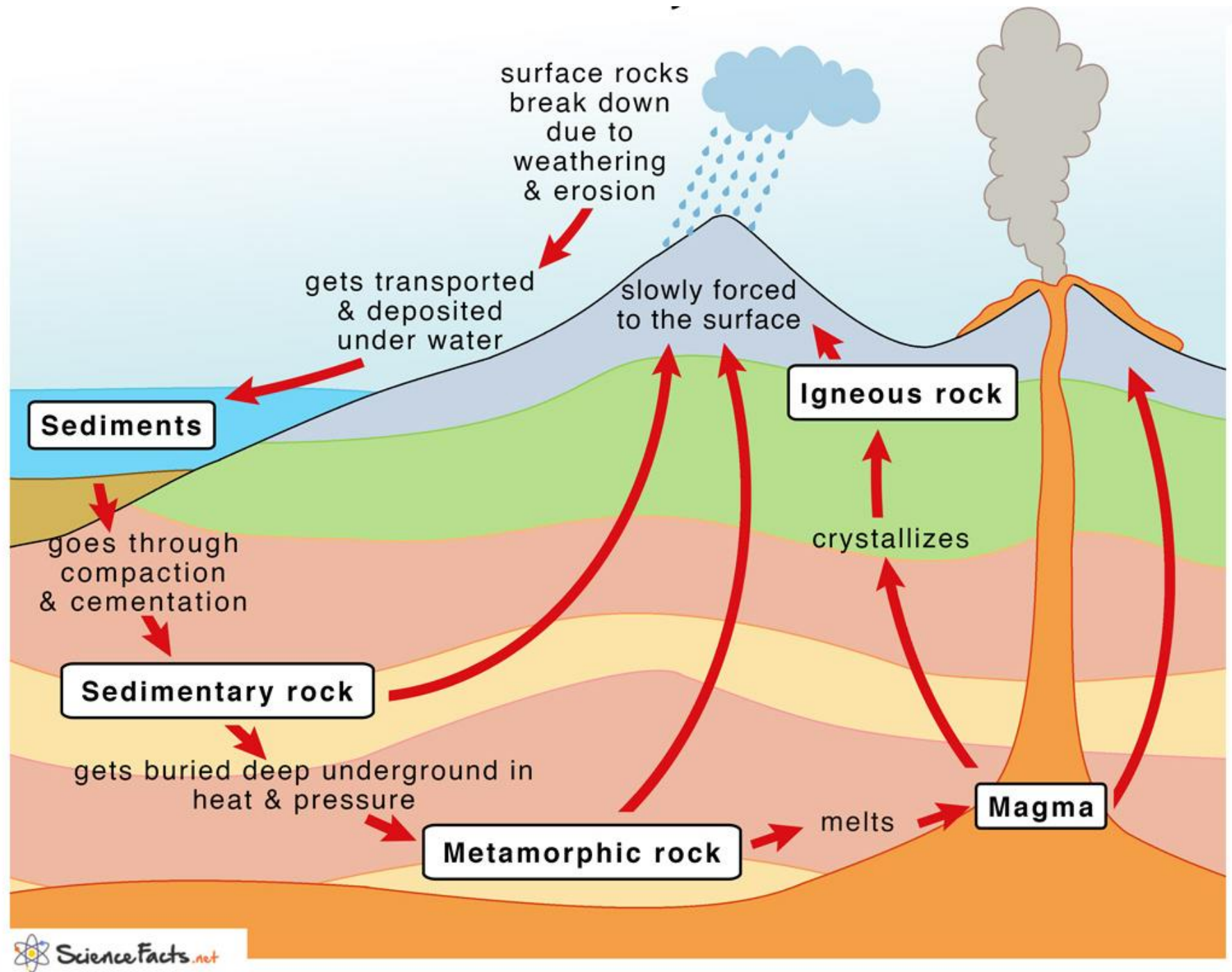
Transport

Deposition

Burial

Subduction

Uplift



Soil's role in the biogeochemical cycles

- Most active area or hotspots for cycling (via both abiotic and biotic processes) of different elements in the lithosphere
- Not only provide nutrients and water for terrestrial ecosystems, but also store and exchange gases with the atmosphere, and affect the movement of surface and groundwater – **interactions/transfers with biosphere, atmosphere and hydrosphere.**
- Material transformations and interactions that occur between soil, sediment, rock, water and the atmosphere during uplift and erosion are important in global biogeochemical cycles.

Source and Sink; Input and output for different elements